

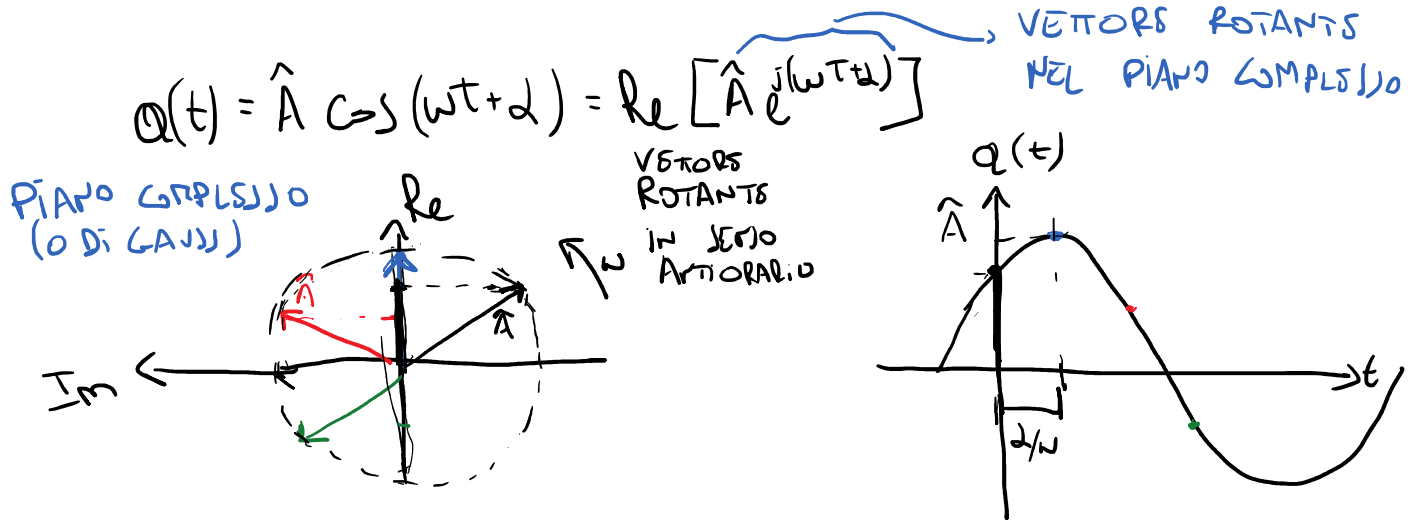
FASORE

mercoledì 20 aprile 2022 18:17

Rappresenteremo una grandezza sinusoidale tramite un numero complesso (un vettore). Questo ci permetterà di risolvere i circuiti in maniera più agevole.

$$\text{OPERATORS COMPLESSO} = \sqrt{-1}$$

$$\text{IDENTITÀ DI EULERO: } e^{jx} = \cos(x) + j \sin(x)$$



SEGNALI ISOFREQU.

$$\begin{aligned} Q_1 &= \hat{A}_1 \cos(\omega t + \alpha_1) \longleftrightarrow \hat{A}_1 e^{j(\omega t + \alpha_1)} = \hat{A}_1 e^{j\omega t} e^{j\alpha_1} = \sqrt{2} A_1 e^{j\omega t} e^{j\alpha_1} \\ Q_2 &= \hat{A}_2 \cos(\omega t + \alpha_2) \longleftrightarrow \hat{A}_2 e^{j(\omega t + \alpha_2)} = \hat{A}_2 e^{j\omega t} e^{j\alpha_2} = \sqrt{2} A_2 e^{j\omega t} e^{j\alpha_2} \\ &\vdots \\ Q_n &= \hat{A}_n \cos(\omega t + \alpha_n) \longleftrightarrow \hat{A}_n e^{j(\omega t + \alpha_n)} = \hat{A}_n e^{j\omega t} e^{j\alpha_n} = \sqrt{2} A_n e^{j\omega t} e^{j\alpha_n} \end{aligned}$$

FAATORE ROTANTE

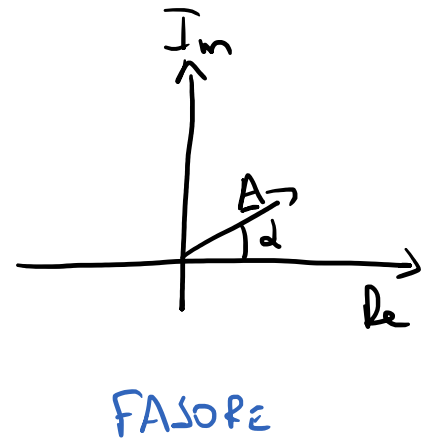
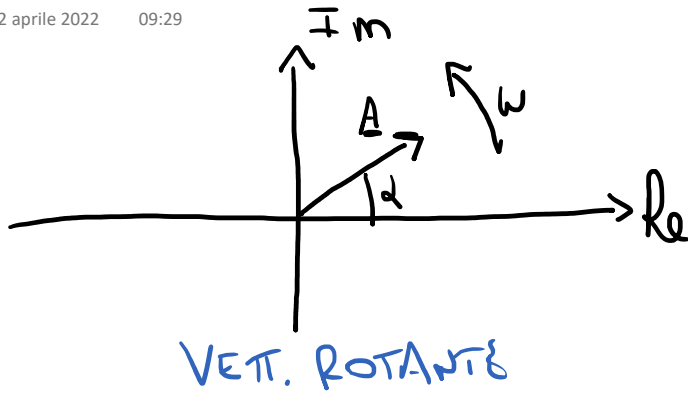
Tutti i vettori ruotano alla stessa pulsazione in senso orario nel piano complesso.

$$\text{Fabbre } \underline{A} = A e^{j\omega t}$$

Annotations:

- FAVORABLE (pointing to the equation)
- VECTORS IN THE COMPLEX PLANE (pointing to $\underline{A}$)
- VAL. EFF. (pointing to A)

Il metodo Fabbre vale per segnali isofrequenziali



TRASFORMATA DI STEINMETZ S[]

$$S[a(t)] = \frac{\sqrt{2}}{T} \int_0^T a(t) e^{-j\omega t} dt$$

$$a(t) = \hat{A} \cos(\omega t + \alpha)$$

$$\rightarrow S[a(t)] = \frac{\sqrt{2}}{T} \int_0^T \hat{A} \cos(\omega t + \alpha) e^{-j\omega t} dt$$

$$= \frac{\sqrt{2}}{T} \hat{A} \int_0^T \frac{e^{j(\omega t + \alpha)} + e^{-j(\omega t + \alpha)}}{2} \cdot e^{-j\omega t} dt =$$

$$= \frac{\sqrt{2}}{T} \frac{\hat{A}}{2} \int_0^T [e^{j(\omega t + \alpha)} + e^{-j(\omega t + \alpha)}] \cdot e^{-j\omega t} dt =$$

$$= \frac{\sqrt{2}}{T} \frac{\hat{A}}{2} \left[\int_0^T e^{j\alpha} dt + \int_0^T e^{-2j\omega t + j\alpha} dt \right] =$$

$$= \frac{\hat{A}}{\sqrt{2}} \cdot e^{j\alpha} \cancel{T} = \frac{\hat{A}}{\sqrt{2}} e^{j\alpha} = \underline{A e^{j\alpha}}$$

FASORE

Associa ad una grandezza nel tempo una grandezza nel dominio Fasioreale

La Trasformata $S[\]$ è un operatore lineare:

$$S[m a(t) + n b(t)] = m S[a(t)] + n S[b(t)]$$

$$S\left[\frac{da(t)}{dt}\right] = \frac{\sqrt{2}}{T} \int_0^T \frac{da(t)}{dt} e^{-j\omega t} dt$$

integrando per parti:

$$S\left[\frac{da(t)}{dt}\right] = \frac{\sqrt{2}}{T} \left[\cancel{[a(t) e^{j\omega t}]_0^T} - \int_0^T a(t) (-j\omega e^{j\omega t}) dt \right]$$

$$= \frac{\sqrt{2}}{T} \left[+j\omega \int_0^T a(t) e^{j\omega t} dt \right] =$$

$$= j\omega \underbrace{\frac{\sqrt{2}}{T} \int_0^T a(t) e^{j\omega t} dt}_{S[a(t)]}$$

$$S[a(t)]$$

$$\boxed{S\left[\frac{da(t)}{dt}\right] = j\omega S[a(t)]}$$

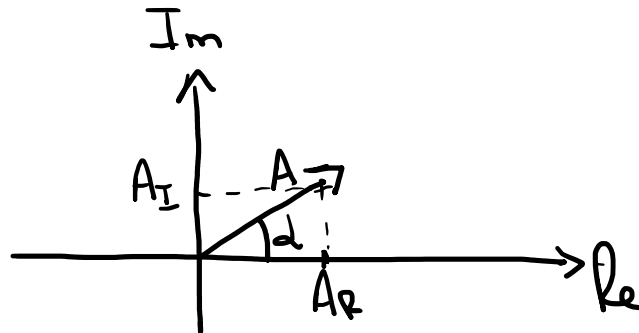
↑
SFASAMENTO

OPERAZIONI SUI NUMERI COMPLESSI

venerdì 22 aprile 2022 09:56

È una rappresentazione del Fasore

Fasore $\begin{cases} \rightarrow \text{Rappres. cartesiana: } \underline{A} = A_R + j A_I \\ \rightarrow \text{Rappres. polare: } \underline{A} = A e^{j\alpha} \end{cases}$

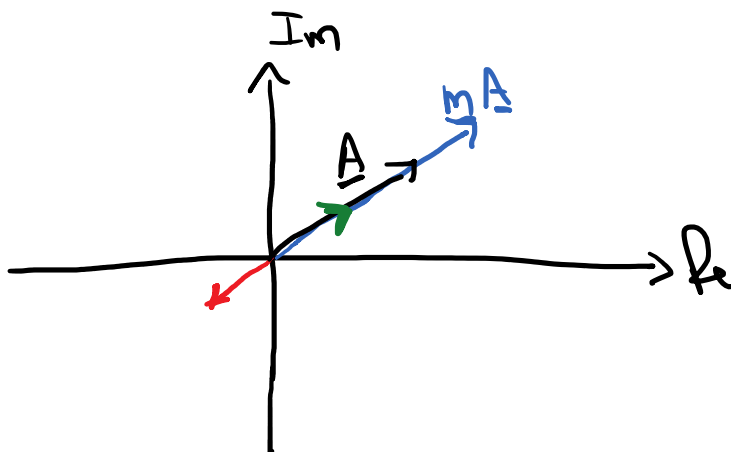


POLARE $\rightarrow$ CARTESIANA

$$A e^{j\alpha} \quad \begin{cases} A_R = A \cos(\alpha) \\ A_I = A \sin(\alpha) \end{cases}$$

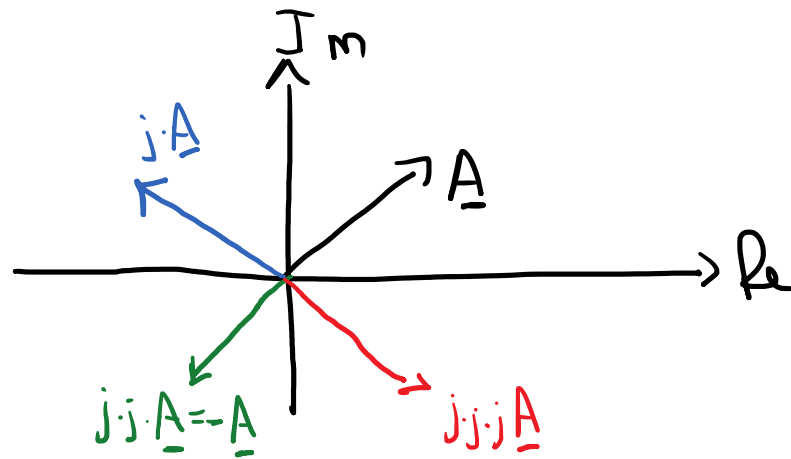
CARTESIANA $\rightarrow$ POLARE

$$A_R + j A_I \quad \begin{cases} A = \sqrt{A_R^2 + A_I^2} \\ \alpha = \arctan\left(\frac{A_I}{A_R}\right) \end{cases}$$



$$m \underline{A} \\ m > 1 \\ 0 < m < 1 \\ m < 0$$

L'effetto di $j \Rightarrow e^{j\frac{\pi}{2}}$



$$\underline{A} = A_R + j A_I = A e^{j\alpha}$$

$$\underline{B} = B_R + j B_I = B e^{j\beta}$$

SOMMA

$$\underline{C} = \underline{A} + \underline{B} = A_R + j A_I + B_R + j B_I = \underbrace{(A_R + B_R)}_{C_R} + j \underbrace{(A_I + B_I)}_{C_I}$$

PRODOTTO

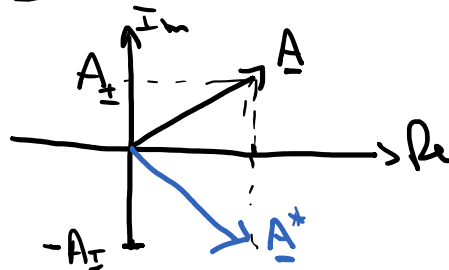
$$\underline{C} = \underline{A} \cdot \underline{B} = A e^{j\alpha} \cdot B e^{j\beta} = \underbrace{A \cdot B}_C e^{j(\alpha + \beta)}$$

RAPPORTO

$$\underline{C} = \frac{\underline{A}}{\underline{B}} = \frac{A e^{j\alpha}}{B e^{j\beta}} = \underbrace{\frac{A}{B}}_C e^{j(\alpha - \beta)}$$

COMPLESSO CONIUGATO

$$\underline{A} = A_R + j A_I \longrightarrow \underline{A}^* = A_R - j A_I = A e^{-j\alpha}$$



METODO SIMBOLICO: Utilizziamo Steinmetz sulle eq. differenziali in modo da avere delle eq. algebriche

$$LKT: \sum_{i=1}^3 v_i(t) = 0 \Rightarrow \sum_{i=1}^3 \underline{V}_i = 0$$

$$LKC: \sum_{i=1}^3 i(t) = 0 \Rightarrow \sum_{i=1}^3 \underline{I}_i = 0$$

EQ. COSTITUTIVE IN FORMA SIMBOLICA:

$$R: v(t) = R i(t) \rightarrow S[v(t)] = \underline{V} = S[R i(t)] = R S[i(t)] \\ \rightarrow \underline{V} = R \underline{I}$$

$$L: v(t) = L \frac{di}{dt} \rightarrow S[v(t)] = \underline{V} = S\left[L \frac{di}{dt}\right] = \\ = L S\left[\frac{di}{dt}\right] = L j\omega S[i(t)] \\ \rightarrow \underline{V} = j\omega L \underline{I}$$

$$C: i(t) = C \frac{dv}{dt} \rightarrow S[i(t)] = S\left[C \frac{dv}{dt}\right] = C S\left[\frac{dv}{dt}\right] \\ \rightarrow \underline{I} = j\omega C \underline{V} \rightarrow \underline{V} = \frac{1}{j\omega C} \underline{I} \\ \underline{V} = -\frac{j}{\omega C} \underline{I}$$

$$e(t) = \sqrt{2} E \cos(\omega t + \alpha) \rightarrow S[e(t)] = \underline{E}$$

$$i(t) = \sqrt{2} I \cos(\omega t + \beta) \rightarrow S[i(t)] = \underline{I}$$

Queste eq. vengono dette in forma simbolica
perché compiono i fasori

Definiamo IMPEDENZA $\underline{Z}$:

$$\underline{Z} = \underline{V} / \underline{I}$$

Ad esempio $\underline{V} = V e^{j\alpha}$, $\underline{I} = I e^{j\beta}$:

$$\underline{Z} = \frac{V e^{j\alpha}}{I e^{j\beta}} = \frac{V}{I} e^{j(\alpha - \beta)}$$

ϕ SFAZZAMENTO: È
LA DIFF. DI FASE
INTRODOTTA DAL
COMPONENTE TRA
TENSIONE E CORR.

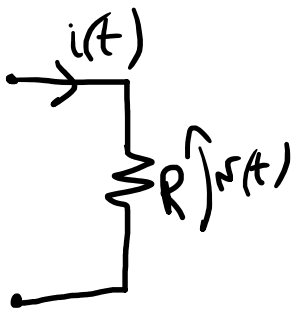
$\phi > 0 \rightarrow \alpha_V > \alpha_I \rightarrow$ tensione è in anticipo rispetto alla corr.

$\phi < 0 \rightarrow \alpha_V < \alpha_I \rightarrow$ tensione è in ritardo rispetto alla corr.

AMPIETUDINE: $\underline{Y} = \frac{1}{\underline{Z}}$

$$\underline{V} = \underline{Z} \underline{I}$$

LEGGE DI OHM
SIMBOLICA

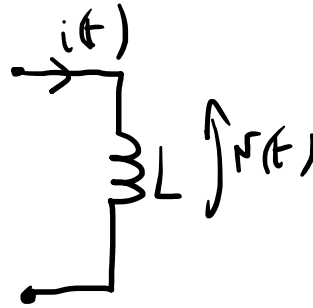
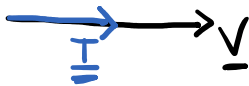


$$\underline{Z} = R$$

$$\underline{I} = \frac{\underline{V}}{\underline{Z}} = \frac{\underline{V}}{R}$$

$$I = \frac{V}{R}$$

$$\alpha_V = 0, \varphi = 0 \rightarrow \alpha_I = 0$$

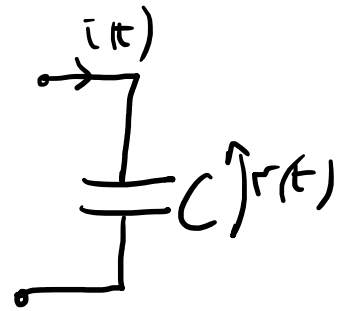


$$\underline{Z} = j\omega L$$

$$\underline{I} = \frac{\underline{V}}{\underline{Z}} = \frac{\underline{V}}{j\omega L}$$

$$I = \frac{V}{\omega L}$$

$$\alpha_V = 0, \alpha_I = -\frac{\pi}{2}$$



$$\underline{Z} = \frac{1}{j\omega C} = -\frac{j}{\omega C}$$

$$\underline{I} = \frac{\underline{V}}{\underline{Z}} = j\omega C \underline{V}$$

$$I = \omega C V$$

$$\alpha_V = 0, \alpha_I = \frac{\pi}{2}$$



EQ. COSTITUTIVE IN FORMA SIMBOLICA

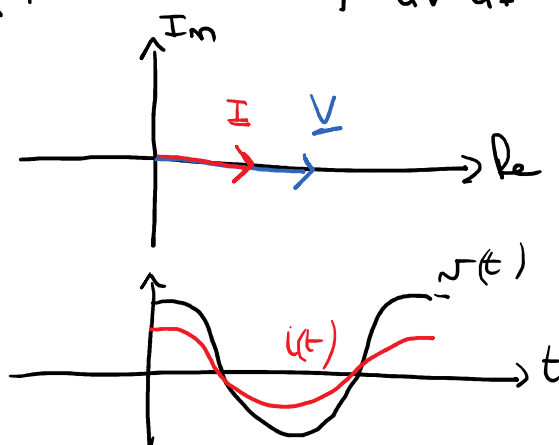
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R)

$$v(t) = R i(t) \xrightarrow{S.I.} \underline{V} = R \underline{I} \rightarrow \underline{Z} = \frac{\underline{V}}{\underline{I}} = R$$

$$\begin{cases} Z_R = R \\ Z_I = 0 \end{cases} \quad \text{È SQUADANTE REALE}$$

$$\begin{cases} |Z| = R \\ \varphi = 0 \end{cases} \quad \begin{array}{l} \text{Supponiamo } \alpha_V = 0 \\ \varphi = \alpha_V - \alpha_I \rightarrow \alpha_I = -\varphi = 0 \end{array}$$

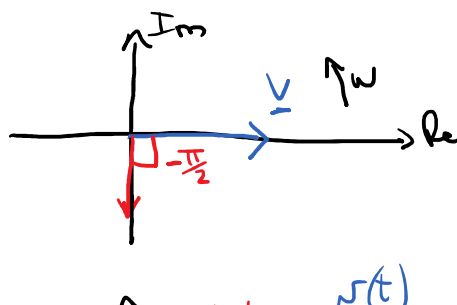


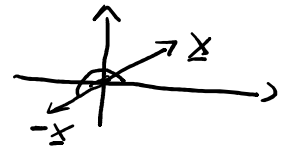
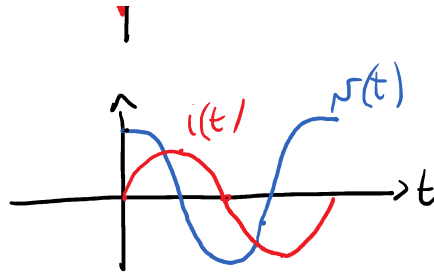
L)

$$v(t) = L \frac{di}{dt} \xrightarrow{S.I.} \underline{V} = j\omega L \underline{I} \rightarrow \underline{Z} = \frac{\underline{V}}{\underline{I}} = j\omega L = \omega L e^{j\frac{\pi}{2}}$$

$$\begin{cases} Z_R = 0 \\ Z_I = \omega L \end{cases} \quad \text{SOLO IMMAGINARIA}$$

$$\begin{cases} |Z| = \omega L \\ \varphi = \frac{\pi}{2} \end{cases} \quad \begin{array}{l} \text{Supponiamo } \alpha_V = 0 \\ \varphi = \alpha_V - \alpha_I \rightarrow \alpha_I = -\varphi = -\frac{\pi}{2} \end{array}$$





c)

$$i(t) = C \frac{dv}{dt} \xrightarrow{SCT} \underline{I} = j\omega C \underline{V} \rightarrow \underline{Z} = \frac{\underline{V}}{\underline{I}} = \frac{1}{j\omega C} \cdot \frac{j}{j} = -\frac{j}{\omega C} = \frac{1}{\omega C} e^{-j\pi/2} = \frac{1}{\omega C} e^{j\pi/2}$$

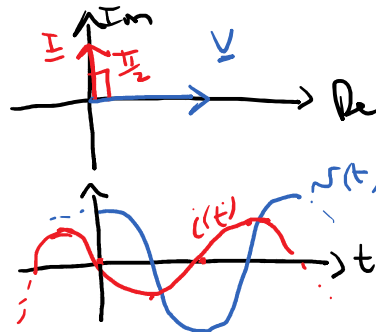
$$\begin{cases} Z_R = 0 \\ Z_I = -\frac{1}{\omega C} \end{cases}$$

SOLO PARTE IMMAGINARIA
NEGATIVA

$$\begin{cases} |Z| = \frac{1}{\omega C} \\ \varphi = -\frac{\pi}{2} \end{cases}$$

Supponiamo $\alpha_V = 0$

$$\varphi = \alpha_V - \alpha_I \rightarrow \alpha_I = -\varphi = \frac{\pi}{2}$$



$$\underline{Z} = Z_R + j Z_I$$

$$\downarrow$$

$$R + jX$$

REATTANZA

$$\rightarrow X_L = \omega L$$

REATTANZA INDUTTIVA

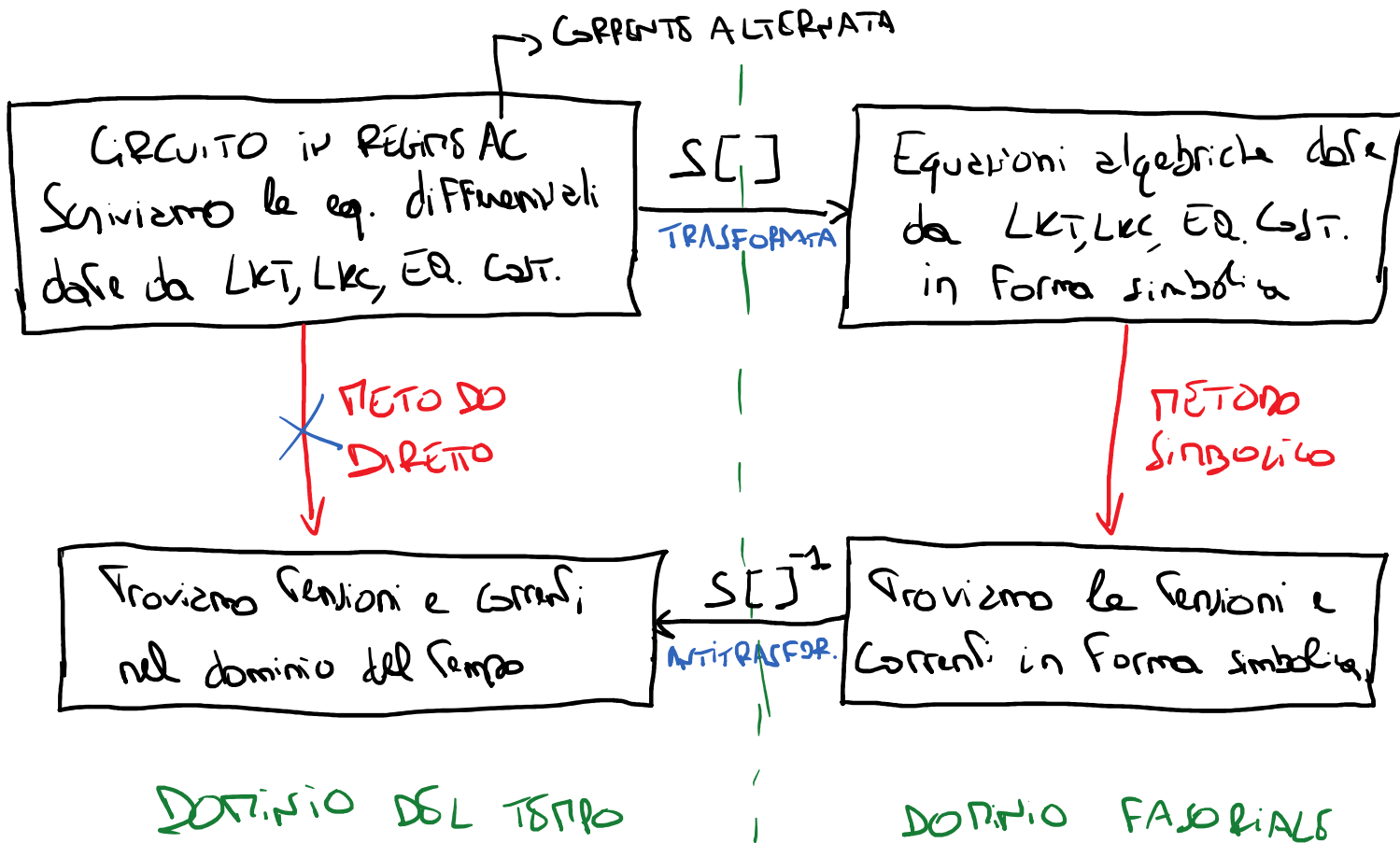
$$\rightarrow X_C = -\frac{1}{\omega C}$$

REATTANZA CAPACITIVA

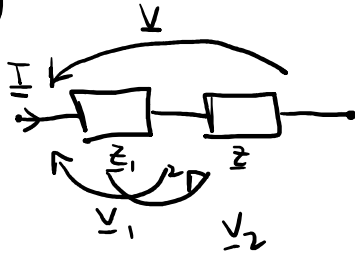
$X = f(\omega) \rightarrow$ RISPOSTA IN FREQ.

METODO SIMBOLICO

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SERIES



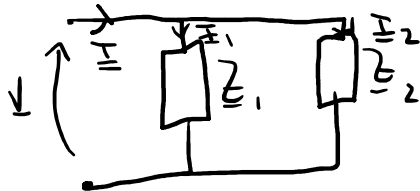
$$\begin{aligned} V &= V_1 + V_2 = Z_1 I + Z_2 I = \\ &= \underbrace{(Z_1 + Z_2)}_{Z_{eq}} I \end{aligned}$$

$$Z_1 = Z_{R1} + j Z_{I1}$$

$$Z_2 = Z_{R2} + j Z_{I2}$$

$$\Rightarrow Z_{eq} = (Z_{R1} + Z_{R2}) + j (Z_{I1} + Z_{I2})$$

PARALLEL



$$\begin{aligned} I &= I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2} = \underbrace{\left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)}_{\frac{1}{Z_{eq}} = Y_{eq}} V \end{aligned}$$